

# The Remarkable Efficiency of the Probabilistic Serial Mechanism

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Is the choice of allocation mechanism irrelevant in the random assignment problem? Existing results conflict: unlike Probabilistic Serial (PS), Random Priority (RP) is almost never ordinally efficient in markets where each object has only one copy, yet the two mechanisms are asymptotically equivalent in replica economies as the number of copies goes to infinity. Whether the one-copy divergence is quantitatively meaningful, and what happens when the number of copies grows slowly, were open questions that we resolve.

With  $q$  copies of  $m$  object types and  $mq$  agents with i.i.d. uniform preferences, we show that PS produces a near-constant average rank with high probability, and its expected average rank is at most  $\log^* m + 3.4$  for every  $q$ . Here  $\log^*$  is the iterated logarithm, a function which grows so slowly that it is below 5 whenever  $m$  is smaller than the number of atoms in the universe. On the other hand, RP's expected average rank is heavily dependent on  $q$ . When  $q$  is constant, it grows logarithmically, diverging from PS. With increasing  $q$  and  $m$ , the gap between PS's and RP's expected average ranks vanishes if and only if  $q$  grows faster than  $\log m$ .

PS's near-constant efficiency is robust and extends to imbalanced and correlated markets.

**Keywords:** probabilistic serial, random priority, random markets.

## 1 Introduction

The Random Priority mechanism (RP) assigns objects by drawing an order of agents and letting each choose their favourite remaining object. It is a commonly used mechanism, partly because it is strategy-proof and ex-post efficient. Yet, the probability distribution that it generates over assignments need not be ordinally efficient, meaning that agents may want to trade different probability shares over objects. To address RP's ordinal inefficiency, [Bogomolnaia and Moulin \[2001\]](#) proposed a new mechanism called Probabilistic Serial (PS). Under PS, all agents simultaneously consume their favourite available objects at the same speed. The resulting fractional assignment is not only ordinally efficient, but also envy-free, at the cost of losing strategy-proofness.

After Bogomolnaia and Moulin's seminal paper, the literature has tackled the natural follow-up question: does PS's ordinal efficiency guarantee meaningfully better assignments than RP? Two articles give sharply different answers. [Manea \[2009\]](#) showed that, in the original setup with the same number of agents and objects, the fraction of preference profiles for which RP is ordinally efficient vanishes, and thus PS and RP differ with high probability. On the other hand, [Che and Kojima \[2010\]](#) consider replica economies with a fixed set of object types. They show that as the number of object copies grows, the allocations generated by RP and PS are asymptotically equivalent. Thus, they argue, *"it may be misleading to simply examine whether a mechanism suffers ordinal inefficiencies; even if a mechanism is ordinally inefficient, the magnitude of the inefficiency may be very small"*.

At the same time, Che and Kojima note that RP's ordinal inefficiency need not vanish if the number of copies is fixed (even if arbitrarily large), and conclude that the comparison between PS and RP in finite economies remains an open question. Manea similarly notes that the Che and Kojima equivalence *"applies only to environments in which the quota of each object is considerably larger than the number of object types"*, and writes that no bound for the expected inefficiency as a function of the quotas is known. In summary, three questions remained open: i) how large is RP's inefficiency in the original setting with no replicas, ii) how does each mechanism's expected inefficiency depend on the number of copies in finite economies, and iii) how many replicas do we need to close their efficiency gap?

We answer the aforementioned three questions in this paper, assuming in the main text that agents' preferences are drawn independently and uniformly at random.<sup>1</sup> We establish explicit bounds on PS's expected average rank for every finite market size, showing that PS's expected average rank admits a nearly constant upper bound that does not depend on the number of object replicas. Furthermore, we identify how many object copies are needed to close the efficiency gap between the two mechanisms.

To be precise, we focus on random markets with  $m$  object types with  $q$  copies each, and  $mq$  agents with preferences drawn independently and uniformly at random. We measure efficiency by the average rank generated by each mechanism under truthful preference revelation, with lower ranks meaning better allocations.<sup>2</sup> Average ranks are widely used as welfare statistics, as they measure how good the objects agents receive are, with each object's rank weighted by its assignment probability, based purely on ordinal information without any assumptions on preference intensity. **Theorem 1** shows that PS achieves near-constant average rank, both in expectation and with high probability, regardless of the number of object copies in the economy. Its expected average rank is at most  $\log^* m + 3.4$ , where  $\log^*$  is the number of times the natural logarithm must be applied before the result is at most one. The iterated logarithm grows extraordinarily slowly: for example,  $\log^* m = 4$  even when  $m = 10^{80}$ , the estimated number of atoms in the universe [Cormen et al., 1990]. In the no-replicas setting ( $q = 1$ ), RP's expected average rank grows logarithmically, which means the gap between the mechanisms diverges. For example, with 10,000 objects with no replicas, PS's expected average rank is at most 6.4, whereas RP's is about 8.8 (see **Figure 1** for an illustration).

**Theorem 2** then identifies exactly when object replication closes the PS-RP efficiency gap. The gap remains if the number of copies is fixed. As the number of object types grows, RP catches up with PS in expected average rank if and only if copies per type grow faster than the logarithm of the number of types. When copies grow more slowly than this logarithm, the gap diverges. When they grow proportionally to it, RP's expected rank becomes bounded, but a positive gap from

<sup>1</sup>**Appendix C.3** extends the results allowing for a class of correlated preferences drawn according to objects' popularity scores.

<sup>2</sup>In our i.i.d. setup, PS is ordinal Bayesian incentive compatible [Dasgupta and Mishra, 2022].

PS persists. [Theorem 2](#) provides a resolution to the tension in the literature. In the one-copy setting studied by Manea, the rank gap diverges. With the fixed set of types considered by Che and Kojima, increasing the number of copies makes RP catch up. Convergence comes from RP improving as larger capacities make demand fluctuations small relative to supply.

The intuition for PS's upper bound is as follows. When few object types remain available, agents may have to consume goods they rank poorly. PS performs well because this part of the process is short. All agents continue eating until time one, so the surviving types must serve the entire population. We show that most of these types have many consumers and therefore exhaust their remaining supply quickly.

The challenge is to establish this demand bound for the types that actually survive, since their survival depends on agents' earlier choices. We prove that the bound holds throughout the process with high probability, and use it to divide consumption into an iterated-logarithmic number of phases, each contributing at most a constant to average rank.

*Further Literature.* After Bogomolnaia and Moulin, much has been written about PS and its properties. Yet, only a handful of papers have studied the quality of assignments that PS generates in practice. [Hugh-Jones et al. \[2014\]](#) conduct an experiment comparing PS to RP in the lab in small markets with only 3 agents and 3 objects. They find evidence of PS being more efficient than RP under observed non-truthful behaviour in one of two relevant treatments. For markets where each object has many copies, [Pathak \[2006\]](#) compares PS with RP using submitted preferences from over eight thousand students in the 2003-04 supplementary round of New York City's high school match. The PS lottery stochastically dominates the RP lottery for 50% of students, while the reverse holds for 6%. Nevertheless, the differences in assignment probabilities are small for most students. [Kornbluth and Kushnir \[2025\]](#) study a course allocation problem with roughly six thousand students and seven hundred fifty courses and find that, among students whose schedules change, PS increases mean estimated utility by 31.8% for first-year students, while reducing it by 3.3%, 16.9%, and 8.6% for second-, third-, and fourth-year students, respectively.

In a recent paper, [Eraslan et al. \[2026\]](#) compare instead PS to RP when assigning roughly nine hundred students to high schools in Seattle. They also estimate cardinal preferences over schools, and with the estimated utilities find that both mechanisms generate almost identical average normalized expected utility.

With regards to the incentive properties of PS, there is a wide theoretical literature with mixed conclusions. The closest paper to us in this literature is [Dasgupta and Mishra \[2022\]](#), who show that if all agents have the uniform prior (as in the i.i.d. uniform model), PS is ordinal Bayesian incentive compatible. [Kojima and Manea \[2010\]](#) is also related to our work, showing that truthful preference reporting becomes a weakly dominant strategy in large replica markets. On the negative side, PS may yield equilibrium outcomes that are neither ordinally efficient nor envy free in specific markets [\[Ekici and Kesten, 2016\]](#). Yet, computing a best response in PS is computationally complex, and can only yield a modest utility gain versus truthful reporting [\[Aziz et al., 2015, Huang et al., 2024\]](#). These results are helpful to explain why most subjects who manipulate PS in practice end up with lower payoffs [\[Hugh-Jones et al., 2014\]](#). In their experiment, roughly 20% and 27% of reports are insincere under RP and PS, respectively, and misreporting is significantly more frequent under PS than under RP in only one of the eight decision situations.

*Random Markets.* There is an extensive literature analyzing the performance of resource allocation algorithms on average rather than on specific instances. This literature, starting with seminal contributions from computer scientists to settle the performance of the Gale-Shapley deferred acceptance procedure in one-to-one markets [\[Knuth, 1976, Wilson, 1972\]](#), has been extended to analyze the performance of a wider family of allocation mechanisms under a variety of market conditions [\[Ashlagi et al., 2017, Che and Tercieux, 2017, Frieze and Pittel, 1995, Knuth, 1996, Ortega and Klein, 2023, Ortega et al., 2026, Pittel, 1989, 1992, Pritchard and Wilson, 2023\]](#).

A recurring theme in this literature is that different mechanisms become equivalent as markets grow large. For example, [Che and Tercieux \[2018\]](#) establish asymptotic payoff equivalence among Pareto-efficient mechanisms under random

preferences, and show that their *normalized* rank distributions also converge.<sup>3</sup> Relatedly, Liu and Pycia [2016] establish asymptotic allocation equivalence among regular mechanisms that are symmetric, asymptotically efficient, and asymptotically strategy-proof, while keeping the set of object types fixed. Our results characterize the comparison between PS and RP when the number of types also grows. We identify exactly when their unnormalized expected average ranks converge: copies per type must grow faster than the logarithm of the number of types.

*Outline.* Section 2 introduces the model and states the upper bounds for PS. Section 3 compares PS with RP and states the quota threshold for convergence. Section 4 proves the high-probability bound in Theorem 1 and explains the refinement giving its explicit expectation bound. Section 5 presents the extensions to unequal quotas, unbalanced markets and correlated preferences. Appendix A contains the auxiliary PS arguments, Appendix B proves the RP comparison, and Appendix C proves the extensions.

## 2 Model and PS's Average Rank

We use  $I$  and  $G$  to denote sets of  $n$  agents and  $m$  types of objects, respectively.<sup>4</sup> Each type has  $q \geq 1$  identical copies and  $n = mq$ . Each agent demands one unit of shares of potentially different types. Agent  $i$  has a strict preference order  $\succ_i$  over the set of object types  $G$ . A preference profile is denoted by  $\succ = (\succ_i)_{i \in I}$ . A market is a tuple  $(I, G, q, \succ)$ .

For each object type  $g \in G$ , let

$$\text{rank}_i(g) := 1 + |\{h \in G : h \succ_i g\}| \quad (1)$$

denote the position of object type  $g$  in agent  $i$ 's preference list, so that rank 1 is the most preferred object type, and rank  $m$  is the worst. Every agent has an

<sup>3</sup>In the one-copy setting, both PS and RP average ranks divided by market size converge to zero, even though the gap between them diverges.

<sup>4</sup>We use the following asymptotic notation. For positive sequences  $a_n$  and  $b_n$ , we write  $a_n = O(b_n)$  if there exists a constant  $C < \infty$  such that  $a_n \leq Cb_n$  for all sufficiently large  $n$ , and  $a_n = o(b_n)$  if  $a_n/b_n \rightarrow 0$ . Subscripts on  $O$  and  $\Theta$  indicate the fixed parameters on which the constants may depend.  $\Theta$  denotes bounds above and below by positive constant multiples.

independently drawn preference, uniform over the  $m!$  strict preference orders over  $G$ .

A random assignment is a non-negative  $n \times m$  matrix

$$x = (x_{ig})_{i \in I, g \in G}, \quad (2)$$

where  $x_{ig}$  is the probability share of good  $g$  assigned to agent  $i$ . Each row sums to one and each column sums to  $q$ : every agent receives one unit in total, and the full supply of every type is allocated.

A mechanism  $M$  maps each preference profile into a random assignment. Under Probabilistic Serial (PS), all agents eat continuously at unit speed. At each time, every agent eats her favourite object with positive remaining supply. When an object is exhausted, each of its current consumers immediately moves to her favourite surviving good. Under Random Priority (RP), a uniform order of agents is drawn, and each agent chooses one copy of her favourite object type with positive remaining supply.

For  $M \in \{\text{PS}, \text{RP}\}$ , let

$$M(\succ) = (M_{ig}(\succ))_{i \in I, g \in G} \quad (3)$$

denote the assignment produced by mechanism  $M$  at profile  $\succ$ .

For a fixed preference profile, the average rank of the assignment produced by mechanism  $M$  is

$$\text{Rank}(M; \succ) := \frac{1}{n} \sum_{i \in I} \sum_{g \in G} \text{rank}_i(g) M_{ig}(\succ). \quad (4)$$

Let  $\succ$  denote the random preference profile. We write

$$\text{Rank}(M) := \text{Rank}(M; \succ) \quad (5)$$

for the resulting random variable. Expectations and probabilities are taken over  $\succ$ . For RP,  $\text{RP}(\succ)$  already averages over the random priority order, so the only remaining randomness in  $\text{Rank}(\text{RP})$  comes from preferences. These quantities depend on  $m$  and  $q$ , with  $n = mq$ . We suppress these subscripts throughout to lighten notation.

Let  $\log^* x$  denote the iterated logarithm of  $x$ , i.e. the least integer  $k \geq 0$  such that applying the natural logarithm  $k$  times to  $x$  yields a value at most one. [Theorem 1](#) gives an explicit bound on PS's expected average rank and an almost-constant bound with high probability.<sup>5</sup>

**THEOREM 1.** *For all positive integers  $m$  and  $q$ , with  $n = mq$ ,*

$$\mathbb{E} \text{Rank}(\text{PS}) \leq \log^* m + 3.4. \quad (6)$$

*Moreover, the same iterated-logarithmic order holds with high probability: there are constants  $C, c > 0$  such that:*

$$\mathbb{P}(\text{Rank}(\text{PS}) > C(1 + \log^* m)) \leq Cn^{-3} + e^{-cn}. \quad (7)$$

We postpone the proof to [Section 4](#).

### 3 Comparison with Random Priority

The following lemma describes how RP's expected average rank depends on the number of types and their quotas.

**LEMMA 1.** *For all integers  $m \geq 2$  and  $q \geq 1$ ,*

$$\mathbb{E} \text{Rank}(\text{RP}) = \Theta\left(1 + \frac{\log m}{q}\right), \quad (8)$$

*with constants independent of  $m$  and  $q$ . For  $q = 1$  (so  $n = m$ ), [Knuth \[1996\]](#) shows*

$$\mathbb{E} \text{Rank}(\text{RP}) = \frac{m+1}{m}(H_{m+1} - 1) = \log m + O(1), \quad (9)$$

*where  $H_k := \sum_{r=1}^k 1/r$  is the  $k$ -th Harmonic number.*

The general statement follows from the two-sided bounds in [\(115\)](#), which combine the lower bound of [Nikzad \[2022, Proposition 5.2\]](#) with the upper bound we prove in [Appendix B.2](#).

For  $q = 1$ , [\(6\)](#) and [Lemma 1](#) imply, as  $m \rightarrow \infty$ ,

$$\mathbb{E} \text{Rank}(\text{RP}) - \mathbb{E} \text{Rank}(\text{PS}) \longrightarrow \infty. \quad (10)$$

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<sup>5</sup>All logarithms are natural and  $e$  denotes Euler's number.

With  $q = 1$ , (6) and (9) imply that PS has lower expected average rank than RP for every  $m \geq 911$ .<sup>6</sup> With  $q = 1$  and  $m = n = 10,000$ , PS's expected average rank is at most 6.4, whereas RP's is approximately 8.789.

*The quota threshold.* Theorem 1 shows that PS's almost-constant rank guarantee holds uniformly over the number of copies, both in expectation and with high probability. Replication can nevertheless change its comparison with RP, because RP becomes better as the number of copies grows.

THEOREM 2. Consider any sequence with  $m \rightarrow \infty$  and integer quotas  $q = q(m) \geq 1$ .

(1) If  $q = o(\log m)$ , then

$$\mathbb{E} \text{Rank}(\text{RP}) \sim \frac{\log m}{q}, \quad (11)$$

and

$$\mathbb{E} \text{Rank}(\text{RP}) - \mathbb{E} \text{Rank}(\text{PS}) \rightarrow \infty. \quad (12)$$

(2) If  $q/\log m \rightarrow c \in (0, \infty)$ , let  $r_c > 1$  be the unique solution of

$$r_c - 1 - \log r_c = \frac{1}{c}. \quad (13)$$

Then

$$\mathbb{E} \text{Rank}(\text{RP}) \rightarrow r_c, \quad \limsup_{m \rightarrow \infty} \mathbb{E} \text{Rank}(\text{PS}) \leq 1 + \sqrt{\frac{2}{c}} < r_c. \quad (14)$$

In particular, the lower limit of the expected-rank gap is at least  $r_c - 1 - \sqrt{2/c} > 0$ .

(3) If  $q/\log m \rightarrow \infty$ , then

$$\mathbb{E} \text{Rank}(\text{PS}) \rightarrow 1, \quad \mathbb{E} \text{Rank}(\text{RP}) \rightarrow 1. \quad (15)$$

Consequently, along any such sequence,

$$|\mathbb{E} \text{Rank}(\text{RP}) - \mathbb{E} \text{Rank}(\text{PS})| \rightarrow 0 \iff \frac{q}{\log m} \rightarrow \infty. \quad (16)$$

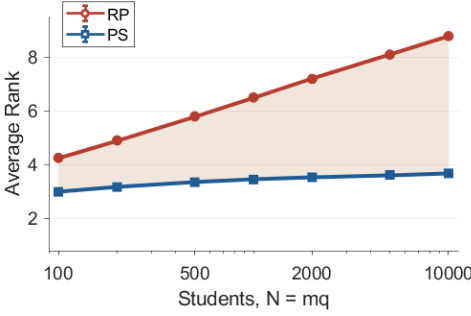
<sup>6</sup>For  $911 \leq m \leq 3.8 \times 10^6$ ,  $\log^* m = 3$ , so PS's expected rank is at most 6.4, whereas RP's expected rank is increasing in  $m$  and exceeds 6.4 at  $m = 911$ . For larger  $m$ , use  $\log^* m + 3.4 \leq 5.4 + \log \log m < \log m - 1 \leq \mathbb{E} \text{Rank}(\text{RP})$ .

The proof is in Appendix B.3. The proof also gives the stronger relative conclusion

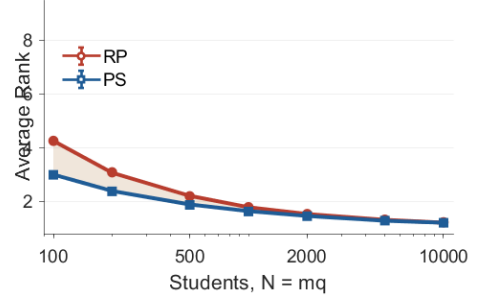
$$\frac{\mathbb{E} \text{Rank}(\text{PS})}{\mathbb{E} \text{Rank}(\text{RP})} \longrightarrow 0 \quad \text{whenever } q = o(\log m). \quad (17)$$

Quotas of order  $\log m$  make RP's expected rank bounded, whereas a vanishing gap from PS requires quotas to grow faster than  $\log m$ .

Figure 1 compares the two sequences at the same total market sizes, starting from 100 types with one copy each. Panel (a) holds  $q = 1$  fixed and increases  $m$ , whereas panel (b) holds  $m = 100$  fixed and increases  $q$ . The first sequence illustrates the widening gap when growth adds distinct alternatives, whereas the second shows RP catching up as both mechanisms improve towards rank one.



(a)  $q = 1, n = m$ .



(b)  $m = 100, q = n/100$ .

Fig. 1. PS and RP under two ways of growing the market. Both panels use  $n = mq \in \{100, 200, 500, 1000, 2000, 5000, 10000\}$  agents and goods and i.i.d. uniform rankings over types. 2,000 markets per  $n$ . The mechanisms use independent samples, and error bars are pointwise 95% Monte Carlo confidence intervals.

*Why RP catches up.* With  $m$  fixed, the number of agents ranking a given type first has mean  $q$  and variance  $q(1 - 1/m)$ . First-choice demand therefore fluctuates on the scale  $\sqrt{q}$ , while capacity grows on the scale  $q$ . Formally, (58) and (115) imply

$$\lim_{q \rightarrow \infty} \mathbb{E} \text{Rank}(\text{PS}) = \lim_{q \rightarrow \infty} \mathbb{E} \text{Rank}(\text{RP}) = 1 \quad \text{for fixed } m.$$

This fixed- $m$  limit uses our uniform preferences and equal quotas. It illustrates the Che–Kojima equivalence in this model, whereas their theorem applies more

generally, including markets in which some types remain disproportionately popular.

## 4 Proof of Theorem 1

Both bounds follow the PS eating process using sequential revelation of preferences. We begin with the high-probability bound (7). The explicit expectation bound (6) is derived in Section 4.4.

### 4.1 Revealing preferences as agents choose

The case  $m = 1$  is immediate, so assume  $m \geq 2$ . Write  $r_g(t) \in [0, q]$  for type  $g$ 's remaining supply, setting it to zero after exhaustion, and define the active set and its size by

$$\mathcal{A}(t) := \{g : r_g(t) > 0\}, \quad A(t) := |\mathcal{A}(t)|. \quad (18)$$

At exhaustion times these quantities refer to the surviving types after the exhausted types have been removed. Treat simultaneous exhaustions as one event, and process the displaced agents' choices sequentially without advancing time. The following deferred-decisions argument, as in Knuth [1976], lets us follow the eating process without conditioning only on its endogenous set of surviving types (the argument is standard in the literature and thus presented without proof).

**LEMMA 2.** *The preference orders can be revealed sequentially so that agents' initial choices are independent and uniform on  $G$ . After any exhaustion event, conditional on the complete history revealed so far, the displaced agents choose independently and uniformly among the goods that remain available.*

*Generating choices by independent draws.* The lemma gives a simple way to generate the choices made during PS. Draw an independent sequence  $\xi_1, \xi_2, \dots$  of uniform labels from  $G$ . Whenever an agent needs to choose, read the next labels until a surviving type appears, skipping labels of exhausted types. The accepted label is uniform among surviving types, so this construction generates the same distribution of choice histories as PS.

For  $S \subseteq G$  and an integer  $k \geq 0$ , let

$$L_S(k) := \sum_{\ell=1}^k \mathbf{1}\{\xi_\ell \in S\}. \quad (19)$$

Let  $K(t)$  be the number of labels read after all displaced agents at time  $t < 1$  have chosen, and let  $c_g(t)$  be the number of agents eating type  $g$ . Every appearance of a type that still survives was accepted and brought a consumer who has not yet left. Hence

$$c_g(t) = L_{\{g\}}(K(t)) \quad (g \in \mathcal{A}(t)), \quad L_{\mathcal{A}(t)}(K(t)) = n. \quad (20)$$

This identity lets us bound demand using counts in an independent sequence. For each fixed set  $S$  and fixed prefix length  $k$ ,  $L_S(k)$  is binomial. The proof below applies standard Chernoff bounds to these counts.

## 4.2 Ranks and demand on surviving goods

At every time between exhaustions, each agent eats her favourite surviving type. For a nonempty set  $S \subseteq G$ , define

$$\rho_i(S) := \min_{g \in S} \text{rank}_i(g). \quad (21)$$

Since  $\text{PS}_{ig}(\succ)$  is the total time agent  $i$  spends eating type  $g$ , changing the order of integration gives

$$\text{Rank}(\text{PS}; \succ) = \int_0^1 \frac{1}{n} \sum_{i=1}^n \rho_i(\mathcal{A}(t)) dt. \quad (22)$$

The finite set of exhaustion times does not affect this integral. We first bound its integrand simultaneously for every possible surviving set.

**LEMMA 3.** *There are constants  $C_0, c_{\text{rank}} > 0$  such that, with probability at least  $1 - e^{-c_{\text{rank}} n}$ ,*

$$\frac{1}{n} \sum_{i=1}^n \rho_i(S) \leq C_0 \frac{m}{|S|} \quad (23)$$

*simultaneously for every nonempty  $S \subseteq G$ .*

The proof is in Appendix A.1.1. Write  $\mathcal{E}_{\text{rank}}$  for the event in this lemma. On this event,

$$\text{Rank}(\text{PS}) \leq C_0 \int_0^1 \frac{m}{A(t)} dt. \quad (24)$$

The integrand is large when few types remain. To bound their contribution, we show that these types attract many eaters and are exhausted quickly. For  $T \subseteq S \subseteq G$ , with  $S$  nonempty, define

$$D(T, S) := \sum_{i=1}^n \mathbf{1}\{\text{agent } i\text{'s favourite type in } S \text{ belongs to } T\}. \quad (25)$$

Thus  $D(T, \mathcal{A}(t))$  is the number of agents eating types in  $T$  at time  $t$ .

PROPOSITION 1. *There are constants  $\eta_0, \alpha > 0$  and  $\Lambda_0 < \infty$ , independent of  $m$  and  $q$ , with the following property. Define*

$$b(s) := \left\lceil m \exp\left(-\alpha \frac{m}{s}\right) \right\rceil, \quad 1 \leq s \leq m. \quad (26)$$

*With probability at least  $1 - O(n^{-3})$ , simultaneously at every time between exhaustion events with  $A(t) = s \leq m/\Lambda_0$ , fewer than  $b(s)$  active goods are being eaten by fewer than*

$$\frac{\eta_0 n}{4 s} \quad (27)$$

*agents. Consequently, every  $T \subseteq \mathcal{A}(t)$  with  $|T| \geq 2b(s)$  satisfies*

$$D(T, \mathcal{A}(t)) \geq \frac{\eta_0 n |T|}{8 s}. \quad (28)$$

PROOF. Set  $\eta_0 := 1/10$  and require  $\Lambda_0 \geq 2/\eta_0$ . Appendix A.1.2 specifies  $\alpha$  and the remaining conditions on  $\Lambda_0$ .

*Step 1: enough labels must have been read.* For each  $1 \leq s \leq m$ , define the deterministic prefix length

$$k_s := \left\lfloor \eta_0 \frac{nm}{s} \right\rfloor. \quad (29)$$

For a fixed set  $S$  of size  $s$ ,  $L_S(k_s)$  is binomial with mean at most  $\eta_0 n$ . The upper-tail Chernoff bound gives

$$\mathbb{P}(L_S(k_s) \geq n) \leq \exp\left[-n(\log(1/\eta_0) - 1 + \eta_0)\right]. \quad (30)$$

If the mean is zero, this event is impossible. Since  $\log(1/\eta_0) - 1 + \eta_0 > \log 2$  and  $m \leq n$ , a union bound over all nonempty  $S$  shows that, except with probability  $e^{-cn}$  for some  $c > 0$ , every such count is below  $n$ . On this event,

$$A(t) = s \implies K(t) > k_s \quad (0 \leq t < 1). \quad (31)$$

Otherwise, the counting identity (20) would give  $n = L_{\mathcal{A}(t)}(K(t)) \leq L_{\mathcal{A}(t)}(k_s) < n$ .

*Step 2: a surviving set with few consumers is unlikely.* Fix  $s \leq m/\Lambda_0$ , put  $b := b(s)$ , and fix  $B \subseteq G$  of size  $b$ . The condition  $\Lambda_0 \geq 2/\eta_0$  ensures  $k_s \geq \eta_0 nm/(2s)$ , so

$$\mu_B := \mathbb{E}L_B(k_s) = \frac{bk_s}{m} \geq \frac{\eta_0 bn}{2s}. \quad (32)$$

If  $B \subseteq \mathcal{A}(t)$  at some time with  $A(t) = s$  and each type in  $B$  has fewer than  $\eta_0 n/(4s)$  consumers, then on the event (31),

$$L_B(k_s) \leq L_B(K(t)) < \frac{\eta_0 bn}{4s} \leq \frac{\mu_B}{2}. \quad (33)$$

The probability that this happens and (31) holds is therefore at most

$$e^{-\mu_B/8} \leq \exp\left(-\frac{\eta_0}{16} \frac{bn}{s}\right), \quad (34)$$

by the lower-tail Chernoff bound. Appendix A.1.2 takes the union bound over  $B$  and  $s$  and derives (28), completing the proof.  $\square$

### 4.3 Bounding average rank by phases

Let  $\mathcal{E}_{\text{load}}$  denote the event that, at every time between exhaustion events with  $A(t) = s \leq m/\Lambda_0$ , fewer than  $b(s)$  surviving types have fewer than  $\eta_0 n/(4s)$  consumers. By Proposition 1, this event has probability at least  $1 - O(n^{-3})$ . We work on this event throughout the following phase argument. A phase beginning with  $\ell$  active goods ends once at most  $2b(\ell)$  remain. The goods that survive the phase have limited remaining supply, while Proposition 1 ensures that they are eaten quickly. This bounds each phase's contribution to the integral. The number of surviving types then falls fast enough that only an iterated-logarithmic number of phases is needed.

PROPOSITION 2. *On the event in Proposition 1,*

$$\int_0^1 \frac{m}{A(t)} dt = O(\log^* m). \quad (35)$$

PROOF. Increase  $\Lambda_0$ , if necessary, so that

$$\frac{\alpha\lambda}{2} \geq \log(8/\alpha) \quad (36)$$

for every  $\lambda \geq \Lambda_0$ . Increasing  $\Lambda_0$  only narrows the range  $s \leq m/\Lambda_0$ , so the conclusion of Proposition 1 remains valid. Increasing  $\Lambda_0$  further ensures  $2b(\ell) < \ell$  whenever  $3 \leq \ell \leq m/\Lambda_0$ : indeed,  $2b(\ell) \leq 2\ell(m/\ell)e^{-\alpha m/\ell} + 2 \leq \ell/4 + 2 < \ell$ . Thus every nonterminal phase defined below has positive length. Let

$$t_0 := \inf\{t : A(t) \leq m/\Lambda_0\}. \quad (37)$$

Before  $t_0$ ,  $m/A(t) \leq \Lambda_0$ , so the initial segment contributes at most  $\Lambda_0$  to the integral.

*Step 1: bound the contribution of one phase.* Suppose  $t_j < 1$  and set  $\ell_j := A(t_j) > 2$ . A phase beginning at  $t_j$  ends when the number of active goods falls to at most  $2b(\ell_j)$ . Define

$$t_{j+1} := \inf\{t > t_j : A(t) \leq 2b(\ell_j)\}. \quad (38)$$

This time exists because  $A(1) = 0$ . For a positive-length phase, let  $T_j$  be the set of goods that survive until immediately before the phase ends, and put

$$T_j := \mathcal{A}(t_{j+1}-), \quad r_j := |T_j|. \quad (39)$$

Then  $r_j > 2b(\ell_j)$ . For every  $t \in [t_j, t_{j+1})$ , goods can only be exhausted, so  $T_j \subseteq \mathcal{A}(t)$  and  $A(t) \leq \ell_j$ . Because  $s \mapsto m \exp(-\alpha m/s)$  is increasing,

$$r_j > 2b(\ell_j) \geq 2b(A(t)). \quad (40)$$

Applying (28) to  $T_j \subseteq \mathcal{A}(t)$  gives, for almost every time in the phase,

$$D(T_j, \mathcal{A}(t)) \geq \frac{\eta_0}{8} \frac{nr_j}{A(t)}. \quad (41)$$

The goods in  $T_j$  together have remaining supply at most  $qr_j$  at the beginning of the phase. Because every agent eats at unit speed, the amount of these goods consumed

during the phase is  $\int_{t_j}^{t_{j+1}} D(T_j, \mathcal{A}(t)) dt$ . Therefore,

$$qr_j \geq \int_{t_j}^{t_{j+1}} D(T_j, \mathcal{A}(t)) dt \geq \frac{\eta_0 qr_j}{8} \int_{t_j}^{t_{j+1}} \frac{m}{A(t)} dt. \quad (42)$$

Since  $n = mq$ , the factor  $q$  cancels, and every phase contributes at most  $8/\eta_0$  to the integral.

*Step 2: bound the number of phases.* Each phase has bounded contribution, so it remains to bound how many phases occur. Put  $\lambda_j := m/\ell_j$ ; the average number of agents per active type at the beginning of phase  $j$  is  $q\lambda_j$ . Because  $A(t_j) \leq A(t_0) \leq m/\Lambda_0$ , we have  $\lambda_j \geq \Lambda_0$  in every phase. If  $t_{j+1} = 1$ , there is no further phase. Suppose therefore that  $t_{j+1} < 1$ , and write  $\ell_{j+1} := A(t_{j+1})$ . By the definition of  $t_{j+1}$  and right-continuity,

$$\ell_{j+1} \leq 2 \lceil me^{-\alpha\lambda_j} \rceil. \quad (43)$$

If  $me^{-\alpha\lambda_j} < 1$ , then  $\ell_{j+1} \leq 2$  and the construction terminates. Otherwise,

$$\ell_{j+1} \leq 4me^{-\alpha\lambda_j}, \quad \lambda_{j+1} \geq \frac{1}{4}e^{\alpha\lambda_j}. \quad (44)$$

Let  $u_j := \alpha\lambda_j/2$ . By (36),

$$u_{j+1} \geq \frac{\alpha}{8}e^{2u_j} \geq e^{u_j}. \quad (45)$$

Thus  $u_j$  grows at least by repeated exponentiation. The number of iterations before  $u_j \geq (\log m)/2$ , at which point  $me^{-\alpha\lambda_j} \leq 1$ , is at most  $\log^* m + O(1)$ .

*Step 3: bound the terminal segment.* Once at most two goods remain, their total remaining supply is at most  $2q$ . The remaining time is therefore at most  $2q/n = 2/m$ , and the terminal integral contributes at most two. Adding the initial segment, the phase contributions, and the terminal segment proves (35).  $\square$

On  $\mathcal{E}_{\text{rank}} \cap \mathcal{E}_{\text{load}}$ , combine (24) with [Proposition 2](#). Taking constants  $C, c > 0$  independent of  $m$  and  $q$  sufficiently large and small, respectively, gives

$$\mathbb{P}(\text{Rank}(\text{PS}) > C(1 + \log^* m)) \leq Cn^{-3} + e^{-cn}, \quad (46)$$

which proves (7). Since  $\text{Rank}(\text{PS}) \leq m \leq n$ , the exceptional event contributes at most  $Cn^{-2} + ne^{-cn}$  to expected rank. The same argument therefore gives  $\mathbb{E} \text{Rank}(\text{PS}) = O(\log^* m)$ , uniformly over  $q$ , for  $m \geq 2$ .

#### 4.4 Expected-rank bounds

We now sharpen the expectation bound by accounting for how much of each allocation remains when its rank increases.

List all “eating” choices in chronological order, including the  $n$  initial choices at time zero. Choice  $h$  occurs at time  $\tau_h$ , when  $a_h$  object types remain available. Each agent selects any object type at most once, so there are at most  $nm$  choices.

Define the cumulative exposure and its integral by

$$v(t) := q + \sum_{\substack{\text{post-initial choices } h \\ \tau_h \leq t}} \frac{1}{a_h}, \quad C(t) := \int_0^t v(s) ds, \quad f_q(t) := \frac{1}{q} \mathbb{E}C(t). \quad (47)$$

Each summand  $1/a_h$  is the conditional probability that choice  $h$  selects any particular surviving type. The initial  $n$  choices each contribute  $1/m$ , giving  $v(0) = n/m = q$ .

In  $C(t)$ , a choice made at time  $\tau_h \leq t$  contributes  $(t - \tau_h)/a_h$ : its selection probability weighted by the consumption time remaining until  $t$ . The deterministic function  $f_q(t)$  is the expectation of  $C(t)$  divided by capacity. The following lemma shows that its value at time one bounds expected average rank.

LEMMA 4. *The expected average rank under PS satisfies*

$$\mathbb{E} \text{Rank}(\text{PS}) \leq 1 + \left(1 - \frac{1}{m}\right) (f_q(1) - 1) \leq f_q(1). \quad (48)$$

PROOF. Reveal each agent’s preference list only up to her current choice. Suppose the exhausted type has rank  $r$  and  $a$  types remain. All  $a$  surviving types lie in the agent’s unrevealed suffix of length  $m - r$ . Conditional on the revealed history, their positions form a uniform  $a$ -element subset of that suffix. The expected position of the first is  $(m - r + 1)/(a + 1)$ : the selected positions divide the other  $m - r - a$  positions into  $a + 1$  exchangeable gaps. Therefore,

$$\mathbb{E}[\text{rank increase} \mid \text{revealed history}] = \frac{m - r + 1}{a + 1} \leq \frac{m}{a + 1}. \quad (49)$$

The initial consumed rank is one. A rank increase at time  $\tau$  contributes that increase times  $1 - \tau$  to the agent’s average rank. Summing (49) over post-initial choices,

averaging over  $n$  agents, and using  $a \leq m - 1$  for all post-initial choices gives

$$\mathbb{E} \text{Rank}(\text{PS}) - 1 \leq \frac{1}{q} \mathbb{E} \sum_{h>n} \frac{1 - \tau_h}{a_h + 1} \leq \frac{1 - 1/m}{q} \mathbb{E} \sum_{h>n} \frac{1 - \tau_h}{a_h} = \left(1 - \frac{1}{m}\right) (f_q(1) - 1). \quad (50)$$

□

*Average congestion.* The quantity  $f_q(1)$  also has a direct interpretation in terms of the number of consumers at each type:

$$f_q(1) = \frac{1}{nq} \mathbb{E} \int_0^1 \sum_{g \in G} c_g(t)^2 dt. \quad (51)$$

For a fixed eating history,  $\sum_g c_g(t)^2/n$  is the average, across agents, of the number of consumers at their chosen type, including themselves. Thus  $f_q(1)$  is average congestion over the eating process, divided by capacity.

To prove (51), define  $\mathcal{W}(t) := \sum_g r_g(t) c_g(t)$ . Immediately after the initial choices,  $\mathcal{W}(0) = nq$ . Between choices,  $\mathcal{W}'(t) = -\sum_g c_g(t)^2$ . Removing exhausted types leaves  $\mathcal{W}$  unchanged because their remaining supply is zero. Each subsequent choice adds the remaining supply of the selected type, whose conditional expectation is  $n(1 - \tau_h)/a_h$ . Since  $\mathcal{W}(1) = 0$ , summing these changes and taking expectations gives

$$\mathbb{E} \int_0^1 \sum_g c_g(t)^2 dt = nq + n \mathbb{E} \sum_{h>n} \frac{1 - \tau_h}{a_h} = n \mathbb{E} C(1), \quad (52)$$

which proves the identity.

*Remaining supply and the quota bound.* The same quantity  $\mathcal{W}(t)$  gives an exact identity for the remaining supply. Since  $r'_g(t) = -c_g(t)$  almost everywhere and each type's supply falls from  $q$  to zero,

$$\int_0^1 \mathcal{W}(t) dt = -\frac{1}{2} \int_0^1 \frac{d}{dt} \sum_g r_g(t)^2 dt = \frac{mq^2}{2}. \quad (53)$$

Thus every type contributes  $q^2/2$  to this integral, regardless of when its capacity is consumed.

To use this identity, define the squared-weight sum

$$B := \sum_h \frac{(1 - \tau_h)^2}{a_h} = 2 \int_0^1 (1 - t)v(t) dt. \quad (54)$$

The sum includes the initial choices, which contribute  $q$ . A choice made late in the process receives little weight because little of the allocation remains.

For any parameters  $\theta, \eta > 0$ , conditional exponential estimates give

$$\mathbb{E}C(1) \leq q + \frac{\log m}{\theta} + \frac{\theta}{2}\mathbb{E}B. \quad (55)$$

$$\mathbb{E}B \leq \frac{\eta q + 1/2}{1 - e^{-\eta}}. \quad (56)$$

Appendix A.2.1 establishes the exponential estimate, and Appendix B.1 derives these two inequalities. The term  $\log m/\theta$  accounts for the  $m$  possible identities of a type that survives until time one. The term involving  $B$  comes from the quadratic remainder in that estimate. A second exponential estimate, weighted by remaining supplies, combines with (53) to bound  $\mathbb{E}B$  independently of the number of types.

Combining the two inequalities and dividing by  $q$  gives

$$f_q(1) \leq 1 + \frac{\log m}{q\theta} + \theta \frac{\eta/2 + 1/(4q)}{1 - e^{-\eta}} \quad (\theta, \eta > 0). \quad (57)$$

Setting  $\eta = q^{-1/2}$  and minimizing over  $\theta > 0$  yields, by Lemma 4,

$$\mathbb{E} \text{Rank}(\text{PS}) \leq 1 + \left(1 - \frac{1}{m}\right) \sqrt{\frac{2 \log m}{q}} \left(1 + \frac{0.55}{\sqrt{q}}\right). \quad (58)$$

The parameter calculation is in Appendix B.1. This is the PS estimate used in Theorem 2: at large quotas its excess rank is bounded by a term of order  $\sqrt{\log m/q}$ .

*The explicit constant.* The bound uniform over quotas uses the time-dependent estimates for  $f_q(t)$  in Appendix A.2. They bound its increments over phases as the remaining fraction of each allocation falls. For  $m \geq 3$ , the initial segment contributes less than 2.108. Each subsequent phase contributes at most one plus a remainder, and these remainders sum to less than 2.291. At most  $\log^* m - 1$  phases

are needed, including the final one. Thus,

$$\mathbb{E} \text{Rank}(\text{PS}) \leq f_q(1) < 2.108 + (\log^* m - 1) + 2.291 < \log^* m + 3.4. \quad (59)$$

Appendix A.3 verifies the phase construction and these constants. For  $m = 1, 2$ , average rank is at most  $m$ . Together with Section 4.3, this completes Theorem 1.

## 5 Extensions

### 5.1 Unequal quotas

Let each type  $g \in G$  have quota  $q_g \geq 1$ , and let  $n = \sum_{g \in G} q_g$  agents have independent uniform rankings over the  $m$  types. Write  $\bar{q} = n/m$  for the average quota and  $q_{\max} = \max_g q_g$  for the largest quota.

**PROPOSITION 3.** *Fix  $K \geq 1$  and suppose  $q_{\max} \leq K\bar{q}$ . There are constants  $C_K, c_K > 0$ , depending only on  $K$ , such that*

$$\mathbb{P}(\text{Rank}(\text{PS}) > C_K(1 + \log^* m)) \leq C_K n^{-3} + e^{-c_K n}. \quad (60)$$

Consequently,  $\mathbb{E} \text{Rank}(\text{PS}) = O_K(\log^* m)$  as  $m \rightarrow \infty$ .

Expected initial demand per type equals the average quota, while remaining supply is bounded using the largest quota. Their ratio changes the phase bound only by a factor  $K$ . The proof appears in Appendix C.1.

### 5.2 Unequal numbers of agents and goods

For the next two results, let there be  $n$  agents and  $m$  unit-capacity goods, with independent uniform preferences. Under PS, agents eat at unit speed until either every agent has consumed one unit or all goods have been exhausted. Under RP, agents choose in a uniform random order until either all agents have chosen or all goods have been assigned. A feasible assignment is a non-negative  $n \times m$  matrix satisfying

$$\sum_{g \in G} x_{ig} \leq 1 \quad \text{for every } i, \quad \sum_{i \in I} x_{ig} \leq 1 \quad \text{for every } g, \quad \sum_{i \in I} \sum_{g \in G} x_{ig} = \min\{n, m\}. \quad (61)$$

Thus every agent receives one unit when  $m \geq n$ , and all goods are allocated when  $n \geq m$ .

*More goods than agents.* When  $m \geq n$ , average rank retains its original definition in (4).

**THEOREM 3.** Fix  $B < \infty$  and consider any sequence  $n \rightarrow \infty$  with  $n \leq m = m(n) \leq Bn$ . Put

$$\Delta := m - n, \quad \Delta_+ := \Delta \vee 1. \quad (62)$$

Then

$$\mathbb{E} \text{Rank}(\text{PS}) = O_B \left( \log^* \left( 1 + \frac{n}{\Delta_+} \right) \right), \quad (63)$$

whereas

$$\mathbb{E} \text{Rank}(\text{RP}) = \frac{m+1}{n} (H_{m+1} - H_{\Delta+1}). \quad (64)$$

Consequently,

$$\frac{\mathbb{E} \text{Rank}(\text{PS})}{\mathbb{E} \text{Rank}(\text{RP})} \rightarrow 0 \iff m - n = o(n). \quad (65)$$

If  $(m - n)/n \rightarrow c > 0$ , then

$$\mathbb{E} \text{Rank}(\text{RP}) \rightarrow (1 + c) \log \frac{1 + c}{c}. \quad (66)$$

Here  $B$  bounds supply relative to demand, and the PS bound's constant may depend on  $B$ . Thus a growing surplus need not wash away the divergence between PS and RP. A proportional surplus does make RP's expected rank bounded. The proof is in Appendix C.2.1.

*More agents than goods.* When  $n \geq m$ , both mechanisms leave total unassigned mass  $n - m$ . To compare the goods allocated, define average rank per allocated unit by

$$\text{Rank}^{\text{alloc}}(M; \succ) := \frac{1}{m} \sum_{i=1}^n \sum_{g=1}^m \text{rank}_i(g) M_{ig}(\succ). \quad (67)$$

Write  $\text{Rank}^{\text{alloc}}(M)$  for the corresponding random variable when preferences are drawn.

THEOREM 4. For every  $n \geq m \geq 1$ ,

$$\mathbb{E} \text{Rank}^{\text{alloc}}(\text{PS}) \leq \min \left\{ \log^* m + 3.4, 1 + \sqrt{\frac{2m \log m}{n}} + \frac{m \log m}{n} \right\}, \quad (68)$$

whereas

$$\mathbb{E} \text{Rank}^{\text{alloc}}(\text{RP}) = \frac{m+1}{m} (H_{m+1} - 1) = \log m + O(1). \quad (69)$$

Consequently, along every sequence  $n \geq m \rightarrow \infty$ ,

$$\frac{\mathbb{E} \text{Rank}^{\text{alloc}}(\text{PS})}{\mathbb{E} \text{Rank}^{\text{alloc}}(\text{RP})} \rightarrow 0. \quad (70)$$

Moreover, for every fixed  $C > 0$ , the PS bound is  $O_C(1)$  whenever  $n/m \geq C \log m$ ; in particular, for every fixed  $\varepsilon > 0$ , it is  $O_\varepsilon(1)$  whenever  $n \geq m^{1+\varepsilon}$ .

The proof is a time-and-capacity rescaling of the balanced market and appears in Appendix C.2.2.

If instead of measuring the rank of only real objects, we could artificially assign rank  $m+1$  to being unassigned. In that case, average rank becomes

$$\text{Rank}^{\text{out}}(M) = \frac{m}{n} \text{Rank}^{\text{alloc}}(M) + \left(1 - \frac{m}{n}\right) (m+1). \quad (71)$$

As  $m \rightarrow \infty$ , if  $m/n$  stays bounded away from one, the ratio of these per-agent expected ranks tends to one. The common second term is at least a positive constant times  $m$ , whereas the expected assigned-good component is  $O(\log m)$  under RP and  $O(\log^* m)$  under PS. This is why we think average rank per allocated unit is the more informative measure of assignment quality.

### 5.3 Correlated preferences

Return to the one-copy case  $q = 1$ , with  $n = m$  agents and  $m$  unit-capacity goods.

DEFINITION 1. Let  $w = (w_g)_{g \in G}$  be a possibly random vector of positive popularity weights, common to all agents, satisfying almost surely

$$\frac{\max_g w_g}{\min_g w_g} \leq \kappa \quad (72)$$

for a fixed  $\kappa < \infty$ . Conditional on  $w$ , agents draw independent Plackett–Luce rankings: each successive good in a ranking is chosen from the remaining goods with probability proportional to its weight.

Random common weights can therefore correlate preferences across agents. Expectations below include both the common weights and the rankings drawn conditional on them.

THEOREM 5. As  $m \rightarrow \infty$ , under [Definition 1](#),

$$\mathbb{E} \text{Rank}(\text{PS}) = O_\kappa(\log^* m), \quad \mathbb{E} \text{Rank}(\text{RP}) = \Theta_\kappa(\log m). \quad (73)$$

Consequently,

$$\frac{\mathbb{E} \text{Rank}(\text{PS})}{\mathbb{E} \text{Rank}(\text{RP})} \rightarrow 0. \quad (74)$$

The proof appears in [Appendix C.3](#). This result requires the stated restriction on dependence: if all agents share the same uniformly drawn ranking, both mechanisms have average rank  $(m + 1)/2$  by [Proposition 6](#).

## 6 Conclusion

Random assignment involves a choice among incompatible desiderata. Random Priority is strategy-proof but not ordinally efficient, whereas Probabilistic Serial is ordinally efficient but not strategy-proof (and these properties are incompatible if equal treatment of equals is also required). Whether giving up strategy-proofness is worth it has been answered by appeal to large-market equivalences. Our results suggest that the answer critically depends on the market. When objects are many and copies are few relative to the logarithm of the number of types, the ordinal efficiency of Probabilistic Serial translates into markedly better assignments. Only when copies per type vastly exceed that logarithm the choice of mechanism becomes irrelevant.

## Acknowledgements

We are grateful to Hervé Moulin and M. Utku Ünver for helpful conversations on this problem.

## AI disclosure

We used AI to improve our upper bound and optimizing its constants. We have verified the proof and stand by the results. We also used AI assistance to produce a Lean formalization of Theorem 1, which is available at [www.josueortega.com](http://www.josueortega.com), together with the Matlab code for Figure 1.

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## A Auxiliary PS arguments

### A.1 High-probability estimates

We prove the uniform rank and load estimates used in [Section 4.2](#). The deterministic phase argument is in [Section 4.3](#). All constants are independent of  $m$  and  $q$ . These estimates also support the extensions in [Section 5](#).

#### A.1.1 Uniform rank control.

PROOF OF [LEMMA 3](#). Fix  $S$  with  $|S| = s$ . For every integer  $0 \leq r \leq m$ ,

$$\mathbb{P}(\rho_i(S) > r) = \frac{\binom{m-r}{s}}{\binom{m}{s}} \leq \exp\left(-\frac{sr}{m}\right). \quad (75)$$

The tail is zero for  $r > m$ . Set  $Y_i := (s/m)\rho_i(S)$ . Integer rounding in the preceding tail bound implies  $\mathbb{P}(Y_i > y) \leq e e^{-y}$  for every  $y \geq 0$ , and hence

$$\mathbb{E} e^{Y_i/2} = 1 + \frac{1}{2} \int_0^\infty e^{y/2} \mathbb{P}(Y_i > y) dy \leq 1 + e. \quad (76)$$

The variables  $Y_i$  are independent across agents. Therefore, for every  $K > 0$ ,

$$\mathbb{P}\left(\sum_{i=1}^n \rho_i(S) > K \frac{nm}{s}\right) \leq \exp\left[-n \left(\frac{K}{2} - \log(1 + e)\right)\right]. \quad (77)$$

Choose  $K$  so that the exponent dominates  $(\log 2 + 2)n$  and take a union bound over the at most  $2^m \leq 2^n$  nonempty sets  $S$ .  $\square$

*A.1.2 Demand from independent draws.* The two estimates in the main proof of [Proposition 1](#) use the standard binomial Chernoff bounds. If  $X$  is binomial with mean  $\mu > 0$ , then

$$\begin{aligned} \mathbb{P}(X \geq x) &\leq \exp[-x \log(x/\mu) + x - \mu] \quad (x > \mu), \\ \mathbb{P}(X \leq \mu/2) &\leq e^{-\mu/8}. \end{aligned} \quad (78)$$

Both follow from  $\mathbb{E} e^{\theta X} \leq \exp[\mu(e^\theta - 1)]$  and Markov's inequality, using positive and negative  $\theta$ , respectively. For use in the extensions, write

$$\psi(u) := \log(1/u) - 1 + u \quad (0 < u < 1). \quad (79)$$

COMPLETION OF THE PROOF OF **PROPOSITION 1**. Keep  $\eta_0 = 1/10$  as chosen in the main proof, and set

$$\beta := \eta_0/16, \quad \alpha := \beta/8. \quad (80)$$

Choose  $\Lambda_0 \geq \max\{2/\eta_0, 1/\alpha\}$  sufficiently large that, for every  $\lambda \geq \Lambda_0$ ,

$$1 + \alpha\lambda \leq (\beta/2)\lambda, \quad 2\lambda e^{-\alpha\lambda} \leq \frac{1}{4}. \quad (81)$$

These choices retain the floor bound  $k_s \geq \eta_0 nm/(2s)$  used in the main proof for  $s \leq m/\Lambda_0$ .

*Step 3: account for all candidate sets.* Fix  $s \leq m/\Lambda_0$ , put  $b := b(s)$ , and write  $\lambda := m/s$ , so  $n/s = q\lambda$ . Since  $b \geq me^{-\alpha\lambda}$ ,

$$\log(em/b) \leq 1 + \alpha\lambda. \quad (82)$$

Taking a union bound in (34) over all  $B \subseteq G$  of size  $b$  and using (81) gives

$$\binom{m}{b} e^{-\beta qb\lambda} \leq e^{-\beta(q-1/2)b\lambda}. \quad (83)$$

Moreover,

$$b\lambda \geq \frac{\log m}{\alpha}. \quad (84)$$

If  $me^{-\alpha\lambda} \geq 1$ , then  $\lambda \leq (\log m)/\alpha$  and  $m\lambda e^{-\alpha\lambda}$  is decreasing on  $[\Lambda_0, (\log m)/\alpha]$ , with value  $(\log m)/\alpha$  at the right endpoint. Otherwise  $b = 1$  and  $\lambda > (\log m)/\alpha$ . Since  $\beta/\alpha = 8$ , the probability of a failure for fixed  $s$  together with (31) is at most  $m^{-8q+4}$ . Summing over  $s$  gives

$$m^{5-8q} \leq (mq)^{-3} = n^{-3}. \quad (85)$$

The inequality follows from  $3 \log q - 8(q-1) \log m \leq (3 - 8 \log 2)(q-1) \leq 0$  for  $m \geq 2$  and  $q \geq 1$ . Adding the exponentially small failure probability of (31) from the main proof gives total failure probability  $O(n^{-3})$ .

On the complementary event, fewer than  $b(s)$  surviving types have fewer than  $\eta_0 n/(4s)$  consumers. Every  $T \subseteq \mathcal{A}(t)$  with  $|T| \geq 2b(s)$  therefore has at least  $|T|/2$  types with at least this many consumers. Summing their loads proves (28).  $\square$

All binomial bounds concern fixed sets and deterministic prefix lengths. The counting identity makes them apply to every surviving set reached by PS. By

**Lemma 2**, the independent sequence generates the same distribution of choice histories as the original preference model, so the resulting load bound holds in that model as well.

## A.2 The explicit expectation bound

We use the chronological choices  $\tau_h, a_h$  and the quantities  $v, C, f_q$  defined in [Section 4.4](#). We first establish a conditional exponential estimate and then derive the time-dependent bounds used for the explicit constant in (6). The same exponential estimate is used in [Appendix B.1](#) for the quota comparison.

*A.2.1 A conditional exponential bound.* The same conditional Chernoff estimate will be used here for the explicit constant and in [Appendix B.1](#) for the quota comparison. Fix a deterministic horizon  $t \leq 1$ . For each type  $g$ , record choices up to time  $t$  and stop recording when it is exhausted. Let  $X_h^g$  indicate a recorded entry into  $g$ , and let  $p_h^g$  be its conditional probability given the preceding history. This probability is  $1/a_h$  while  $g$  is available and the choice occurs by  $t$ , and zero otherwise. Take any non-negative weights  $z_h$  determined before choice  $h$ , and any  $\theta > 0$ . Then

$$M_g := \exp \left\{ -\theta \sum_h z_h X_h^g + \sum_h p_h^g (1 - e^{-\theta z_h}) \right\}, \quad \mathbb{E} M_g \leq 1. \quad (86)$$

This is the usual exponential calculation behind the Chernoff bound, applied by successive conditioning. At each choice, put  $d_h := 1 - e^{-\theta z_h}$ . The conditional expectation of the new factor is  $e^{p_h^g d_h} (1 - p_h^g d_h) \leq 1$ . Multiplying these factors and conditioning successively proves (86); at most  $nm$  choices are made. Pad the sequence to  $nm$  choices with zero terms when necessary.

For the explicit expectation bound, take  $z_h = t - \tau_h$  for choices made by  $t$ , and zero otherwise. For every type still available at  $t$ , the compensator in (86) is the

same quantity

$$\begin{aligned}
 F_{\theta,t} &:= q(1 - e^{-\theta t}) + \sum_{\substack{h>n \\ \tau_h \leq t}} \frac{1 - e^{-\theta(t-\tau_h)}}{a_h} \\
 &= \theta \int_0^t e^{-\theta(t-s)} v(s) ds = \theta C(t) - \theta^2 \int_0^t e^{-\theta(t-s)} C(s) ds.
 \end{aligned} \tag{87}$$

For a surviving type, its weighted entry total is precisely its consumed capacity  $q - r_g(t)$ . Since  $\sum_g r_g(t) = mq(1 - t)$ , the mean consumed capacity across the  $A := A(t)$  surviving types is  $q(1 - m(1 - t)/A)$ . Jensen's inequality gives

$$\sum_g M_g \geq A \exp \left\{ F_{\theta,t} - q\theta \left( 1 - \frac{m(1 - t)}{A} \right) \right\}. \tag{88}$$

Consequently, with  $u := 1 - t > 0$ ,

$$\mathbb{E} F_{\theta,t} \leq \Phi(q\theta, u), \quad \Phi(\lambda, u) := \max_{u \leq a \leq 1} \left\{ -\log a + \lambda \left( 1 - \frac{u}{a} \right) \right\}. \tag{89}$$

To see this without conditioning on the endogenous surviving set, take logs in the preceding display, bound the term depending on  $A/m$  by its maximum, and use  $\mathbb{E} \log(\sum_g M_g/m) \leq 0$ . Writing  $\ell := -\log u$ , the maximum is explicit:

$$\Phi(\lambda, u) = \begin{cases} \ell, & 0 < \lambda \leq 1, \\ \ell + \lambda - 1 - \log \lambda, & 1 \leq \lambda \leq 1/u, \\ \lambda(1 - u), & \lambda \geq 1/u. \end{cases} \tag{90}$$

At  $t = 1$ , at least one type is exhausted exactly at time one. For any such type its weighted entry total is  $q$  and its compensator is  $F_{\theta,1}$ . Thus  $e^{F_{\theta,1} - q\theta} \leq \sum_g M_g$ , which yields

$$f_q(1) \leq 1 + \frac{\log m}{q\theta} + \theta \int_0^1 e^{-\theta(1-s)} f_q(s) ds. \tag{91}$$

Summing the exponential quantities over all types allows for the random identity of the last surviving type.

By (87)–(89),

$$f_q(t) \leq \frac{\Phi(q\theta, 1 - t)}{q\theta} + \theta \int_0^t e^{-\theta(t-s)} f_q(s) ds. \tag{92}$$

For fixed  $u$ ,  $\Phi(\lambda, u)/\lambda$  is nonincreasing in  $\lambda$ . Therefore replacing its first term by  $\Phi(\theta, 1-t)/\theta$  gives a valid inequality simultaneously for every  $q \geq 1$ .

We also need an initial upper bound for the analytical argument. For  $\eta > 0$ , concavity gives  $1 - e^{-\eta z} \geq z(1 - e^{-\eta t})/t$  for  $0 \leq z \leq t$ . Hence  $F_{\eta, t} \geq [(1 - e^{-\eta t})/t]C(t)$ . Apply (89), use the preceding monotonicity in  $q$ , and choose  $\eta = 1 + t$ . The result is

$$f_q(t) \leq \Psi(t) := \frac{t [t - \log(1 - t^2)]}{1 - e^{-t-t^2}}, \quad 0 < t < 1. \quad (93)$$

**A.2.2 The explicit iterated-logarithmic bound.** The next lemma applies the preceding inequalities to deterministic time intervals. Here the phases bound increments of  $f_q$ . Their construction and the verification of the constants are in Appendix A.3.

**LEMMA 5.** *For  $m \geq 3$  and  $q \geq 1$ , put  $z_j := (5/6)2^{-j}$  for  $j \geq 0$ . There are an integer  $J \geq 0$  and times  $0 < s_0 < \dots < s_J < 1$  such that  $J + 1 \leq \log^* m - 1$ ,*

$$\begin{aligned} f_q(s_0) &< 2.108, \\ f_q(s_{j+1}) - f_q(s_j) &\leq e^{z_j}, \quad 0 \leq j < J, \\ f_q(1) - f_q(s_J) &\leq e^{z_J}. \end{aligned} \quad (94)$$

Moreover,  $\sum_{j=0}^{\infty} (e^{z_j} - 1) < 2.291$ .

### A.3 Verification of the explicit phase bounds

**PROOF OF LEMMA 5.** Since the exposure  $v$  is non-decreasing,  $f_q$  is non-decreasing and convex, with  $f_q(0) = 0$  and  $\mathbb{E} \text{Rank}(\text{PS}) \leq f_q(1)$ . We use (92), replacing its first term by  $\Phi(\theta, 1-t)/\theta$ , and (91), replacing  $\log m/(q\theta)$  by  $\log m/\theta$ . These replacements are valid simultaneously for all  $q \geq 1$ .

*A bound for one phase.* Fix  $0 \leq s < t < 1$  and put  $\Delta := t - s$ . Before  $s$ , use  $f_q(r) \leq f_q(s)$ ; on  $[s, t]$ , use the secant bound

$$f_q(r) \leq f_q(s) + \frac{r-s}{\Delta} (f_q(t) - f_q(s)). \quad (95)$$

Substituting these bounds into (92) gives

$$\frac{1 - e^{-\theta\Delta}}{\theta\Delta} (f_q(t) - f_q(s)) \leq \frac{\Phi(\theta, 1-t)}{\theta} - e^{-\theta t} f_q(s) \leq \frac{\Phi(\theta, 1-t)}{\theta}. \quad (96)$$

Consequently,

$$f_q(t) - f_q(s) \leq \frac{\Delta\Phi(\theta, 1-t)}{1 - e^{-\theta\Delta}}. \quad (97)$$

The same calculation applied to (91) yields, for  $u \in (0, 1]$ ,

$$f_q(1) - f_q(1-u) \leq \frac{\theta u + u \log m}{1 - e^{-\theta u}}. \quad (98)$$

We will also use the initial envelope  $\Psi$  in (93).

We construct the times and verify the bounds in Lemma 5.

*Bounding each phase's contribution.* Define

$$z_j := \frac{5}{6} 2^{-j}, \quad c_j := e^{z_j} - 1 - z_j, \quad \lambda_0 := \frac{e}{c_0}, \quad \lambda_{j+1} := e z_j \lambda_j \exp(c_j \lambda_j). \quad (99)$$

Write  $x_j := c_j \lambda_j$ ,  $u_j := 1/\lambda_j$ , and  $s_j := 1 - u_j$ . We first check that

$$x_j \geq e 2^j, \quad x_{j+1} \geq e^{x_j}. \quad (100)$$

Indeed, if  $c(z) := e^z - 1 - z$ , then

$$c(z) = z^2 \int_0^1 (1-r) e^{zr} dr, \quad \frac{c(z/2)}{c(z)} \geq \frac{1}{4} e^{-z/2}. \quad (101)$$

Thus, assuming  $x_j \geq e 2^j$ ,

$$\begin{aligned} x_{j+1} &= \frac{c_{j+1}}{c_j} e z_j x_j e^{x_j} \\ &\geq \frac{1}{4} e^{-5/12} e \frac{5}{6} 2^{-j} (e 2^j) e^{x_j} = \frac{5}{24} e^{19/12} e^{x_j} \geq e^{x_j}. \end{aligned} \quad (102)$$

Since  $e^x \geq 2x$  for  $x > 0$  and  $x_0 = e$ , this proves (100) by induction. It also implies that  $\lambda_j$  increases and  $0 < s_j < s_{j+1} < 1$ .

For phase  $j$ , choose  $\theta_j := z_j \lambda_j$ . Since  $z_j x_j \geq 5e/6$  and  $c_j \leq c_0$ , we have  $\theta_j = z_j x_j / c_j \geq 5e/(6c_0) > 1$ . The definition of  $\lambda_{j+1}$  gives

$$\theta_j u_{j+1} = e^{-1-x_j} < 1. \quad (103)$$

Hence the maximizer in  $\Phi(\theta_j, u_{j+1})$  is the interior point  $a = \theta_j u_{j+1}$ , so

$$\Phi(\theta_j, u_{j+1}) = \theta_j - 1 - \log(\theta_j u_{j+1}) = \theta_j + x_j = \frac{z_j + c_j}{u_j}. \quad (104)$$

The function  $d/(1 - e^{-\theta d})$  increases in  $d > 0$ . Since  $s_{j+1} - s_j \leq u_j$ , (97) therefore yields

$$f_q(s_{j+1}) - f_q(s_j) \leq \frac{z_j + c_j}{1 - e^{-z_j}} = e^{z_j}. \quad (105)$$

*Stopping the phases.* For  $m \geq 3$ , let  $J$  be the first index with  $x_J \geq \log m$ . Apply (98) with  $u = u_J$  and  $\theta = z_J/u_J$ . Since  $u_J \log m \leq c_J$ ,

$$f_q(1) - f_q(s_J) \leq \frac{z_J + c_J}{1 - e^{-z_J}} = e^{z_J}. \quad (106)$$

For the number of phases, let  $E_0 := 1$  and  $E_{k+1} := e^{E_k}$ . Equation (100) implies  $x_j \geq E_{j+1}$ . If  $k := \log^* m$ , then  $k \geq 2$  and  $\log m \leq E_{k-1}$ . Consequently  $J \leq k - 2$ , or  $J + 1 \leq \log^* m - 1$ . This gives the required number of phases.

All constants are explicit. Put  $z := 5/6$ . For the series,

$$\sum_{j=0}^{\infty} (e^{z2^{-j}} - 1) = \sum_{k=1}^{\infty} \frac{z^k}{k!(1 - 2^{-k})} \leq 2z + \frac{4}{3}(e^z - 1 - z) < 2.291. \quad (107)$$

For the initial value,

$$s_0 = 1 - \frac{e^{5/6} - 1 - 5/6}{e}, \quad \Psi(s_0) < 2.108. \quad (108)$$

The last bound follows, for example, from  $.82796 < s_0 < .82797$ : evaluate the numerator of  $\Psi$  at  $.82797$  and its denominator at  $.82796$ . This proves the bounds on the initial segment and the sum of remainders.  $\square$

## B Proofs for the comparison with Random Priority

### B.1 The PS bound with quotas

We derive (55) and (56) from the conditional exponential estimate (86) and the supply identity (53), and then deduce (58). The case  $m = 1$  is immediate, so assume  $m \geq 2$ .

*PROOF. Step 1: weight choices by the allocation still outstanding.* Use the chronological choices and  $v, C, f_q$  defined in (47), and the squared-weight sum  $B$  in (54). Recall that  $C(1) = \sum_h (1 - \tau_h)/a_h$ , including the initial choices. The squared weights in  $B$  arise from the second-order bound  $1 - e^{-x} \geq x - x^2/2$ . Apply (86) at time

one with  $z_h = 1 - \tau_h$ . For any type exhausted at time one, the weighted number of recorded entries is its consumed capacity  $q$ , and its conditional selection probability at every choice is  $1/a_h$ . Consequently,

$$\exp\left\{\theta(C(1) - q) - \frac{\theta^2 B}{2}\right\} \leq \sum_{g \in G} M_g. \quad (109)$$

Taking logarithms and expectations, and using Jensen's inequality and  $\mathbb{E} \sum_g M_g \leq m$ , gives (55). The factor  $m$  accounts for the possible identities of the last surviving type.

*Step 2: use remaining supply to bound B.* Fix  $t < 1$  and  $\eta > 0$ , and put  $u := 1 - t$  and  $b := 1 - e^{-\eta}$ . Apply (86) to choices made by  $t$  with unit weights and parameter  $\eta$ , and denote the resulting quantities by  $\tilde{M}_g(t)$ . For surviving types,

$$\tilde{M}_g(t) = \exp\{bv(t) - \eta c_g(t)\}. \quad (110)$$

After a type exhausts, its recorded entry count stays fixed; the formula above is used only for surviving types. Recall  $\mathcal{W}(t) = \sum_g r_g(t) c_g(t)$ . Since  $r_g(t) \leq q$  and  $\sum_g r_g(t) = mqu$ , weighted Jensen gives

$$\begin{aligned} \sum_g \tilde{M}_g(t) &\geq \frac{1}{q} \sum_g r_g(t) e^{bv(t) - \eta c_g(t)} \\ &\geq mu \exp\left\{bv(t) - \frac{\eta \mathcal{W}(t)}{mqu}\right\}. \end{aligned} \quad (111)$$

Taking logarithms and expectations, using  $\mathbb{E} \sum_g \tilde{M}_g(t) \leq m$ , yields

$$b\mathbb{E}v(t) \leq \frac{\eta \mathbb{E}\mathcal{W}(t)}{mqu} - \log u. \quad (112)$$

Multiply (112) by  $2u$  and integrate. The supply identity (53) and  $\int_0^1 u(-\log u) du = 1/4$  give (56). Combining it with (55) and dividing by  $q$  proves (57).

*Step 3: choose the two parameters.* Set  $\eta := q^{-1/2}$ , write  $x := \eta \leq 1$ , and put  $C_q := (x/2 + x^2/4)/(1 - e^{-x})$ . The elementary inequality  $(1 - e^{-x})^{-1} \leq x^{-1} + 1/2 + x/12$  gives

$$2C_q \leq 1 + x + x^2/3 + x^3/24 \leq (1 + 0.55x)^2. \quad (113)$$

Minimizing  $\log m/(q\theta) + C_q\theta$  over  $\theta > 0$  in (57) yields

$$f_q(1) - 1 \leq \sqrt{2 \log m/q} (1 + 0.55/\sqrt{q}). \quad (114)$$

**Lemma 4** proves (58). For completeness, the reciprocal inequality follows after multiplication from  $(x^2 - 6x + 12)e^x \geq x^2 + 6x + 12$ : the coefficients through degree four cancel, and the coefficient of  $x^k$  for  $k \geq 5$  is  $(k - 3)(k - 4)/k! \geq 0$ .  $\square$

## B.2 RP ranks and quotas

LEMMA 6. For every  $m \geq 2$  and  $q \geq 1$ ,

$$\max \left\{ 1, \frac{\mathbb{E} \text{Rank}(\text{RP})|_{q=1}}{q} \right\} \leq \mathbb{E} \text{Rank}(\text{RP}) \leq 1 + \sqrt{\frac{2 \log m}{q}} + \frac{\log m}{q}. \quad (115)$$

where  $\mathbb{E} \text{Rank}(\text{RP})|_{q=1}$  denotes the expected average rank with one copy of each type. For every fixed  $m$ ,  $\mathbb{E} \text{Rank}(\text{RP})$  is non-increasing in the integer quota  $q$ .

PROOF. The lower bound follows from Nikzad [2022, Proposition 5.2, proof] and  $\mathbb{E} \text{Rank}(\text{RP}) \geq 1$ . For the upper bound and monotonicity, we use an auxiliary clock representation of RP. At each step, the next agent's ranking is independent of the earlier choices. If  $a$  types remain available, her favourite among them is uniformly distributed, and its expected rank is  $(m+1)/(a+1)$ . We can therefore reproduce the distribution of RP's sequence of chosen types by giving each type an independent rate-one Poisson clock: the next ring selects uniformly among the active clocks, uses one copy of that type, and stops its clock if this was the last copy. Only the order of the rings matters for the allocation.

Let  $G_j$  be the time of the  $q$ th ring of clock  $j$ . These times are independent  $\text{Gamma}(q, 1)$  variables, each with mean  $q$  and variance  $q$ . Writing  $F_q$  for their distribution function, the number of active clocks is  $A(t) = |\{j : G_j > t\}| \sim \text{Bin}(m, 1 - F_q(t))$ . When  $A(t) = a$ , allocations occur at rate  $a$ , each with expected rank  $(m+1)/(a+1)$ . Integrating the expected sum of ranks over this auxiliary

time and evaluating the binomial expectation gives

$$\begin{aligned}\mathbb{E} \text{Rank}(\text{RP}) &= \frac{m+1}{mq} \int_0^\infty \mathbb{E} \left[ \frac{A(t)}{A(t)+1} \right] dt \\ &= \frac{1}{mq} \sum_{k=1}^m \mathbb{E} \max_{j \leq k} G_j.\end{aligned}\tag{116}$$

Indeed, the binomial identity is  $(m+1)\mathbb{E}[A(t)/(A(t)+1)] = \sum_{k=1}^m (1 - F_q(t)^k)$ , and integrating each term gives the corresponding expected maximum. In particular,  $\mathbb{E} \text{Rank}(\text{RP}) \leq q^{-1} \mathbb{E} \max_{j \leq m} G_j$ . Thus the upper bound reduces to controlling the largest of  $m$  sample means, each based on  $q$  independent exponential waiting times. For  $0 < \lambda < 1$ , the gamma moment-generating function and Jensen's inequality yield

$$\log \mathbb{E} e^{\lambda(G_j - q)} \leq \frac{q\lambda^2}{2(1-\lambda)}, \quad \mathbb{E} \text{Rank}(\text{RP}) \leq 1 + \frac{\log m}{q\lambda} + \frac{\lambda}{2(1-\lambda)}.\tag{117}$$

Taking  $\lambda = a/(1+a)$  with  $a = \sqrt{2 \log m/q}$  proves the upper bound for  $m \geq 2$ ;  $m = 1$  is immediate. Since  $\mathbb{E} \text{Rank}(\text{RP})|_{q=1} = \log m + O(1)$  by [Lemma 1](#), these bounds give  $\mathbb{E} \text{Rank}(\text{RP}) \sim (\log m)/q$  when  $q = o(\log m)$ , and  $\mathbb{E} \text{Rank}(\text{RP}) \rightarrow 1$  whenever  $(\log m)/q \rightarrow 0$ . Finally,  $G_j/q$  is a mean of  $q$  independent exponentials. A vector of  $(q+1)$ -sample means is the average of its leave-one-out vectors of  $q$ -sample means. Convexity of the maximum and [\(116\)](#) therefore give monotonicity in  $q$ .  $\square$

### B.3 Proof of [Theorem 2](#)

PROOF. We consider the three regimes in turn. If  $q = o(\log m)$ , set  $a = \log m/q \rightarrow \infty$ . By [Lemma 6](#),  $\mathbb{E} \text{Rank}(\text{RP}) \sim a$ , whereas [\(58\)](#) gives  $\mathbb{E} \text{Rank}(\text{PS}) \leq 1 + 1.55\sqrt{2a} = o(a)$ . This proves [\(12\)](#).

Next suppose  $q/\log m \rightarrow c \in (0, \infty)$ . Write  $I(r) = r - 1 - \log r$  for  $r \geq 1$ . For an upper bound, the gamma representation [\(116\)](#) and the gamma moment-generating function give, for  $0 < \lambda < 1$ ,

$$\mathbb{E} \text{Rank}(\text{RP}) \leq \frac{1}{q} \mathbb{E} \max_{j \leq m} G_j \leq \frac{\log m/q - \log(1-\lambda)}{\lambda}.\tag{118}$$

Let  $r_m > 1$  solve  $\mathcal{I}(r_m) = \log m/q$  and choose  $\lambda = 1 - 1/r_m$ . The last expression equals  $r_m$ , which tends to  $r_c$ .

For a lower bound, fix  $y > 1$ . Integrating the  $\text{Gamma}(q, 1)$  density over  $[qy, qy+1]$  and applying Stirling's formula gives

$$\mathbb{P}(G_1 \geq qy) \geq \exp\{-q\mathcal{I}(y) - O(\log q)\} \quad (q \rightarrow \infty). \quad (119)$$

Fix  $\delta \in (0, 1)$  and put  $k = \lfloor \delta m \rfloor$ . For  $1 < y < r_c$ , we have  $c\mathcal{I}(y) < 1$ , and hence

$$\log\{k\mathbb{P}(G_1 \geq qy)\} \geq \log m - q\mathcal{I}(y) - O(\log q) + O(1) \rightarrow \infty. \quad (120)$$

Independence gives  $\mathbb{P}(\max_{j \leq k} G_j \geq qy) \rightarrow 1$ . The summands in (116) are non-decreasing in  $k$ , so

$$\liminf_{m \rightarrow \infty} \mathbb{E} \text{Rank}(\text{RP}) \geq (1 - \delta)y. \quad (121)$$

Letting  $y \uparrow r_c$  and  $\delta \downarrow 0$  proves  $\mathbb{E} \text{Rank}(\text{RP}) \rightarrow r_c$ . Since  $q \rightarrow \infty$ , (58) gives  $\limsup_{m \rightarrow \infty} \mathbb{E} \text{Rank}(\text{PS}) \leq 1 + \sqrt{2/c}$ . For every  $s > 0$ ,  $\mathcal{I}(1+s) = s - \log(1+s) < s^2/2$ : the difference  $s^2/2 - s + \log(1+s)$  is zero at zero and has derivative  $s^2/(1+s) > 0$ . Taking  $s = \sqrt{2/c}$  and using strict monotonicity of  $\mathcal{I}$  on  $(1, \infty)$  proves (14). In particular,

$$\liminf_{m \rightarrow \infty} (\mathbb{E} \text{Rank}(\text{RP}) - \mathbb{E} \text{Rank}(\text{PS})) \geq r_c - 1 - \sqrt{2/c} > 0. \quad (122)$$

Finally, if  $q/\log m \rightarrow \infty$ , (58) and (115), together with the lower bound one, give (15). This proves sufficiency in (16). For necessity, if  $q/\log m$  does not tend to infinity, there is a bounded subsequence and hence a further subsequence converging to some  $c \in [0, \infty)$ . If  $c = 0$ , the rank gap diverges by (12); if  $c > 0$ , its lower limit is strictly positive by (14). Thus the absolute difference of expected ranks cannot converge to zero.  $\square$

## C Proofs of the extensions

### C.1 Unequal quotas

PROOF OF PROPOSITION 3. Write  $\bar{q} := n/m$  and  $q_{\max} := \max_g q_g \leq K\bar{q}$ . The uniform rank and sequential-revelation arguments are unchanged. The sampling construction in the proof of Proposition 1 also applies: the number of appearances

of a surviving type still equals its number of consumers, regardless of its quota. Neither binomial estimate uses the individual capacities. Thus the same load bound holds with  $\bar{q}$  in place of  $q$ , including the failure bound  $m^{5-8\bar{q}} \leq n^{-3}$ , since  $\bar{q} \geq 1$ .

In a phase, any  $r$  surviving types have remaining supply at most  $q_{\max}r$ . The demand bound (28) therefore gives

$$\int_{t_j}^{t_{j+1}} \frac{m}{A(t)} dt \leq \frac{8}{\eta_0} \frac{q_{\max}}{\bar{q}} \leq \frac{8K}{\eta_0}. \quad (123)$$

The number of phases and the initial contribution are unchanged, while the terminal contribution is at most  $2K$ . Combining these estimates with (24) proves the claimed high-probability bound. The expectation bound follows because  $\text{Rank}(\text{PS}) \leq m \leq n$  on the exceptional event.  $\square$

## C.2 Unequal numbers of agents and goods

Use the unbalanced market and assignment conventions of Section 5.2. Put  $\tau = \min\{1, m/n\}$ . For  $0 \leq t < \tau$ , all  $n$  agents are eating. Retain  $\mathcal{A}(t)$  for the set of active goods,  $A(t) = |\mathcal{A}(t)|$ , and  $\rho_i(S) = \min_{g \in S} \text{rank}_i(g)$ . We first adapt the two concentration estimates to unequal market sizes.

LEMMA 7. *Fix  $B < \infty$ . There are constants  $C_B, c_{B,\text{rank}} > 0$  such that, whenever  $m \leq Bn$ , with probability at least  $1 - e^{-c_{B,\text{rank}}n}$ ,*

$$\frac{1}{n} \sum_{i=1}^n \rho_i(S) \leq C_B \frac{m}{|S|} \quad (124)$$

*simultaneously for every nonempty  $S \subseteq G$ .*

PROOF. For a fixed set  $S$  of size  $s$ , the exponential-moment estimate in the proof of Lemma 3 applies unchanged to  $(s/m)\rho_i(S)$ . Choosing  $C_B$  sufficiently large gives a failure probability  $e^{-c_{B,\text{rank}}n}$  for that set. There are at most  $2^m \leq 2^{Bn}$  sets, so a union bound proves the simultaneous statement.  $\square$

The next proposition is the unequal-market version of Proposition 1.

PROPOSITION 4. *Fix  $B < \infty$ . There are constants  $\Lambda_B, \eta_B, \alpha_B, c_{B,\text{load}} > 0$  such that, whenever  $m \leq Bn$ , with probability at least  $1 - O_B(m^{-3}) - e^{-c_{B,\text{load}}n}$ , the following holds*

simultaneously at every time  $t < \tau$  between exhaustion events with  $A(t) = s \leq n/\Lambda_B$ . Define

$$b_{n,m}(s) := \left\lceil m \exp\left(-\alpha_B \frac{n}{s}\right) \right\rceil. \quad (125)$$

Fewer than  $b_{n,m}(s)$  active goods are being eaten by fewer than

$$\frac{\eta_B}{4} \frac{n}{s} \quad (126)$$

agents. Consequently, every  $T \subseteq \mathcal{A}(t)$  with  $|T| \geq 2b_{n,m}(s)$  satisfies

$$D(T, \mathcal{A}(t)) \geq \frac{\eta_B}{8} \frac{n|T|}{s}. \quad (127)$$

PROOF. Use the independent uniform stream from the proof of [Proposition 1](#). Its counting identity holds at every time  $t < \tau$ , since all  $n$  agents are still consuming. Choose  $\eta_B \in (0, 1)$  so small that  $\psi(\eta_B) > B \log 2 + 1$ , and for each  $s$  define

$$k_s := \left\lfloor \eta_B \frac{nm}{s} \right\rfloor. \quad (128)$$

For any fixed set  $S$  of size  $s$ , the number  $L_S(k_s)$  is binomial with mean at most  $\eta_B n$ . The upper-tail bound in (78) gives probability at most  $e^{-\psi(\eta_B)n}$  that this count reaches  $n$ . A union bound over at most  $2^m \leq 2^{Bn}$  sets shows that, except with probability  $e^{-c_{B,\text{load}}n}$ , the stream has passed position  $k_s$  whenever  $A(t) = s$ : otherwise the  $s$  surviving goods could not have all  $n$  consumers.

Set  $\beta_B := \eta_B/16$  and  $\alpha_B := \beta_B/8$ . Fix  $s \leq n/\Lambda_B$ , let  $b := b_{n,m}(s)$ , and fix a set  $B_0$  of size  $b$ . Choose  $\Lambda_B \geq 2/\eta_B$ , so  $k_s \geq \eta_B nm/(2s)$ . The binomial count  $L_{B_0}(k_s)$  therefore has mean at least  $\eta_B bn/(2s)$ . On the preceding high-probability event, if  $B_0$  survives when  $A(t) = s$  and each of its goods has fewer than  $\eta_B n/(4s)$  consumers, this count is below half its mean. The lower-tail bound in (78) therefore bounds the probability of this intersection by  $\exp(-\beta_B bn/s)$ .

Write  $\lambda := n/s$ . As in the proof of [Proposition 1](#),

$$\log \binom{m}{b} \leq b(1 + \alpha_B \lambda). \quad (129)$$

Increase  $\Lambda_B \geq 1/\alpha_B$  so that, for every  $\lambda \geq \Lambda_B$ ,

$$1 + \alpha_B \lambda \leq (\beta_B/2)\lambda, \quad 2B\lambda e^{-\alpha_B \lambda} \leq \frac{1}{4}. \quad (130)$$

The second inequality gives  $2b_{n,m}(s) < s$  for  $s \geq 3$  in the stated range. The same elementary estimate as in (84) gives  $b\lambda \geq (\log m)/\alpha_B$ . Hence the union bound over all  $B_0$  for a fixed  $s$  is at most  $m^{-4}$ , and summing over  $s$  gives  $O(m^{-3})$ . Adding the exponentially small failure probability from the first step gives the stated probability bound. Finally, if  $|T| \geq 2b$ , at least  $|T|/2$  goods in  $T$  have at least  $\eta_B n/(4s)$  consumers, which proves (127).  $\square$

### C.2.1 More goods than agents.

PROOF OF THEOREM 3. On the event in Lemma 7,

$$\text{Rank(PS)} \leq C_B \int_0^1 \frac{m}{A(t)} dt \leq BC_B \int_0^1 \frac{n}{A(t)} dt. \quad (131)$$

We bound the last integral on the event in Proposition 4. Let  $t_0 = \inf\{t : A(t) \leq n/\Lambda_B\} \wedge 1$ . The initial segment contributes at most  $\Lambda_B$  to the integral. If  $t_0 = 1$ , the proof is complete. Otherwise set  $\ell_j = A(t_j)$  and, while  $t_j < 1$  and  $\ell_j > 2$ , define

$$v_j := 2b_{n,m}(\ell_j), \quad t_{j+1} := \inf\{t > t_j : A(t) \leq v_j\} \wedge 1. \quad (132)$$

For every positive-length phase let  $T_j = \mathcal{A}(t_{j+1}-)$  and  $r_j = |T_j|$ . Whether the threshold is reached or the horizon is reached first,  $r_j > v_j$ . For all times in the phase,  $T_j \subseteq \mathcal{A}(t)$  and  $r_j > 2b_{n,m}(\ell_j) \geq 2b_{n,m}(A(t))$ . Thus (127) and the capacity of  $T_j$  give

$$\int_{t_j}^{t_{j+1}} \frac{n}{A(t)} dt \leq \frac{8}{\eta_B}. \quad (133)$$

If a threshold is reached before time 1, put  $\lambda_j = n/\ell_j$  and  $\gamma = n/m \geq 1/B$ . Then

$$\lambda_{j+1} \geq \frac{\gamma}{4} e^{\alpha_B \lambda_j} \quad (134)$$

whenever the unrounded threshold is at least one. After increasing  $\Lambda_B$ , the rescaled sequence grows at least by repeated exponentiation. A further threshold phase is impossible once  $me^{-\alpha_B \lambda_j} < \Delta/4$  when  $\Delta \geq 4$ , because then  $2\lceil me^{-\alpha_B \lambda_j} \rceil \leq \Delta$ , whereas  $A(t) \geq \Delta + n(1-t) > \Delta$  for  $t < 1$ . When  $\Delta \leq 3$ , reaching  $me^{-\alpha_B \lambda_j} < 1$  makes the next threshold equal to two, after which only a bounded terminal piece remains. Thus it is enough in every case to reach  $\alpha_B \lambda_j \geq \log(4m/(\Delta \vee 4))$  up to an additive constant. Repeated exponentiation does so in at most  $O_B(\log^*(1 + n/\Delta_+))$

steps. If the construction reaches a state with at most two active goods before time 1, their total remaining supply is at most two, so the remaining time is at most  $2/n$  and the remaining integral is at most two. Hence the integral and therefore the PS rank obey (63). The exceptional events contribute  $o(1)$  because rank is at most  $m \leq Bn$ .

For RP, the  $k$ th agent sees  $s = m - k + 1$  remaining goods. The remaining set is independent of that agent's preference order, so her expected best rank is  $(m + 1)/(s + 1)$ . Averaging over  $k = 1, \dots, n$  gives (64). If  $\Delta = o(n)$ , this expression is of order  $\log(1 + n/\Delta_+)$ , which dominates its iterated logarithm. If  $\Delta/n$  does not tend to zero, RP's expected rank is bounded along a subsequence while PS's rank is at least one, so the ratio cannot tend to zero. The final limit follows from harmonic-number asymptotics.  $\square$

### C.2.2 More agents than goods.

PROOF OF THEOREM 4. PS exhausts all goods at time  $m/n$ . Multiplying time and every capacity by  $n/m$  turns this market into a replica market with  $m$  types, capacity  $q = n/m$  for each type, and  $n$  agents eating on  $[0, 1]$ . All allocated shares and the total allocated mass are multiplied by the same factor, so average rank per allocated unit equals the replica market's average rank.

The expectation bounds in (6) and (58) also hold for real capacities  $q \geq 1$ , since their proofs do not require  $q$  to be an integer. Substituting  $q = n/m$  and using  $0.55\sqrt{2\log m} \leq \log m$  for  $m \geq 2$  gives (68); the case  $m = 1$  is immediate.

Under RP, only the first  $m$  agents in the priority order receive goods. Their preferences remain independent and uniform, so RP's expected average rank per allocated unit equals that of the balanced market with  $m$  agents and  $m$  goods. This gives (69), and the asymptotic conclusions follow.  $\square$

## C.3 Correlated preferences

Use the one-copy model in Definition 1. Conditional on  $w$ , its rankings can equivalently be generated by independent exponential variables  $Y_{ig} \sim \text{Exp}(w_g)$ , ranked in increasing order. Unless conditioning on  $w$  is stated explicitly, probabilities and

expectations are over both the weights and the conditionally independent rankings. Write  $W(S) = \sum_{g \in S} w_g$  and normalize weights so that  $1 \leq w_g \leq \kappa$ .

LEMMA 8. *There are constants  $\tilde{C}_\kappa, C_\kappa, c_{\kappa, \text{rank}} > 0$  such that, conditional on every admissible weight vector, with probability at least  $1 - e^{-c_{\kappa, \text{rank}} n}$ ,*

$$\frac{1}{n} \sum_{i=1}^n \rho_i(S) \leq \tilde{C}_\kappa \frac{W(G)}{W(S)} \leq C_\kappa \frac{m}{|S|} \quad (135)$$

*simultaneously for every nonempty  $S$ .*

PROOF. Before an agent's exponential race hits  $S$ , the clocks in  $S$  have total rate  $W(S)$  and the total rate of all remaining clocks is at most  $W(G)$ . Thus

$$\mathbb{P}(\rho_i(S) > r \mid w) \leq \exp\left(-r \frac{W(S)}{W(G)}\right). \quad (136)$$

The exponential-moment argument from Lemma 3, followed by a union bound over all  $S$ , proves the first inequality uniformly in  $w$ . The second follows from  $W(G)/W(S) \leq \kappa m/|S|$  after setting  $C_\kappa := \kappa \tilde{C}_\kappa$ .  $\square$

LEMMA 9. *Conditional on the weights and on the complete revealed history, every displaced agent chooses good  $g$  from the current set  $S$  of available goods with probability  $w_g/W(S)$ , independently across displaced agents.*

PROOF. Plackett–Luce rankings have the deletion property: conditional on the previously selected goods, the restriction of the remaining order to any surviving set is again Plackett–Luce with the same weights. Equivalently, in the exponential-race representation, memorylessness leaves independent residual exponential clocks with rates  $(w_g)_{g \in S}$ . Their minimum is  $g$  with probability  $w_g/W(S)$ . Independence across agents is preserved because preferences are revealed only as needed.  $\square$

PROPOSITION 5. *For every fixed  $\kappa < \infty$ , there are constants  $\Lambda_\kappa, \eta_\kappa, \alpha_\kappa, c_{\kappa, \text{load}} > 0$  such that, with probability at least  $1 - O_\kappa(n^{-3}) - e^{-c_{\kappa, \text{load}} n}$ , the following holds simultaneously at every time between exhaustion events with  $A(t) = s \leq m/\Lambda_\kappa$ . Define*

$$b_\kappa(s) := \left\lceil m \exp\left(-\alpha_\kappa \frac{m}{s}\right) \right\rceil. \quad (137)$$

Fewer than  $b_\kappa(s)$  active goods are being eaten by fewer than

$$\frac{\eta_\kappa}{4\kappa} \frac{n}{s} \quad (138)$$

agents. Consequently, every  $T \subseteq \mathcal{A}(t)$  with  $|T| \geq 2b_\kappa(s)$  satisfies

$$D(T, \mathcal{A}(t)) \geq \frac{\eta_\kappa}{8\kappa} \frac{n|T|}{s}. \quad (139)$$

PROOF. Conditional on the weights, use an independent stream in which label  $g$  has probability  $w_g/W(G)$ , skipping exhausted goods. The next surviving label belongs to  $g \in S$  with probability  $w_g/W(S)$ , so Lemma 9 gives the required choice process. The counting identity (20) remains valid: for any surviving set  $S$ , its label count  $L_S(K(t))$  equals its number of consumers.

Choose  $\eta_\kappa > 0$  with  $\kappa\eta_\kappa < 1$  and  $\psi(\kappa\eta_\kappa) > \log 2$ , and define

$$k_s := \left\lfloor \eta_\kappa \frac{nm}{s} \right\rfloor. \quad (140)$$

For a fixed set  $S$  of size  $s$ , its count in this prefix is binomial with mean at most  $\kappa\eta_\kappa n$ , since  $W(S)/W(G) \leq \kappa s/m$ . The upper-tail bound in (78) and a union bound over all  $S$  show that, except with probability  $e^{-c_{\kappa, \text{load}} n}$ , the stream has passed position  $k_s$  whenever  $A(t) = s$ . Here  $n = m$ , and all estimates are uniform over the admissible weights.

Fix  $s \leq m/\Lambda_\kappa$ , put  $b := b_\kappa(s)$ , and fix a set  $B$  of size  $b$ . Choose  $\Lambda_\kappa \geq 2/\eta_\kappa$ . Since  $W(B)/W(G) \geq b/(\kappa m)$ , its count in the first  $k_s$  labels has mean at least  $\eta_\kappa bn/(2\kappa s)$ . On the preceding high-probability event, if  $B$  survives with every good having fewer than  $\eta_\kappa n/(4\kappa s)$  consumers, that prefix count is below half its mean. The lower-tail bound in (78) bounds the probability of this intersection by  $\exp(-\beta_\kappa bn/s)$ , where  $\beta_\kappa := \eta_\kappa/(16\kappa)$ . Set  $\alpha_\kappa := \beta_\kappa/8$  and increase  $\Lambda_\kappa$  as in Proposition 1. The same union bound over  $B$  and  $s$ , together with the exponentially small failure probability from the first step, gives total failure probability  $O_\kappa(n^{-3})$ . If  $|T| \geq 2b_\kappa(s)$ , at least  $|T|/2$  goods in  $T$  each have at least  $\eta_\kappa n/(4\kappa s)$  consumers, which yields (139).  $\square$

PROOF OF THEOREM 5. On the event in Lemma 8,

$$\text{Rank}(\text{PS}) \leq C_\kappa \int_0^1 \frac{m}{A(t)} dt. \quad (141)$$

Use [Proposition 5](#) and the phase construction from [Proposition 2](#), replacing  $b(s)$  by  $b_\kappa(s)$ . Each phase contributes at most  $8\kappa/\eta_\kappa$ , and  $\lambda_j = m/\ell_j$  grows at least as  $\lambda_{j+1} \geq \frac{1}{4} \exp(\alpha_\kappa \lambda_j)$  after a constant initial segment. Hence there are  $O_\kappa(\log^* m)$  phases and the same order holds in expectation.

For RP, condition on  $w$  and on a set  $S$  of  $s$  goods remaining when an agent chooses. The current agent's exponential clocks are independent of  $S$ . The tail estimate used in [Lemma 8](#) gives

$$\mathbb{E}[\rho_i(S) \mid S, w] \leq 1 + \kappa m/s. \quad (142)$$

For the reverse inequality, if  $s < m/(16\kappa)$  set  $r = \lfloor m/(8\kappa s) \rfloor$ . Throughout the first  $r$  draws, conditional on having missed  $S$ , the remaining total weight is at least  $7m/8$ , while  $W(S) \leq \kappa s$ . A union bound implies  $\mathbb{P}(\rho_i(S) > r \mid S, w) \geq 3/4$ , and hence  $\mathbb{E}[\rho_i(S) \mid S, w] \geq cm/(\kappa s)$ . For larger  $s$ , the same lower order follows from  $\rho_i(S) \geq 1$ . Thus the conditional expectation is  $\Theta_\kappa(m/s)$  uniformly in  $S$  and  $w$ . Averaging over RP positions  $s = m, m-1, \dots, 1$  gives  $\Theta_\kappa(H_m) = \Theta_\kappa(\log m)$ .  $\square$

**PROPOSITION 6.** *Suppose one strict order is drawn uniformly and every agent has that same order. Each agent's marginal preference is uniform, but*

$$\text{Rank(PS)} = \text{Rank(RP)} = \frac{m+1}{2} \quad (143)$$

*for every  $m$ .*

**PROOF.** Under PS, all agents eat the common first choice until it is exhausted, then the common second choice, and so on. Each receives  $1/m$  of every good, so average rank is  $(m+1)/2$ . Under RP, the agent in priority position  $r$  receives the common  $r$ th choice, giving the same average.  $\square$